

SAMBP TR-28/2026

Master Synthesis Report: State-Adaptive Model-Based Protection (SAMBP) Technical Report Series TR-03 through TR-27

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PhD Thesis Supporting Report

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Contents

1	Introduction	2
2	Report Registry	2
3	Arc 1: Core Model-Based Framework (TR-03, TR-04)	4
3.1	SAMBP Inverse Estimation Principle	4
4	Arc 2: System Validation (TR-05 to TR-11)	4
4.1	System Integration and Communication (TR-05 to TR-07)	4
4.2	Robustness Validation (TR-08 to TR-11)	4
5	Arc 3: 87L Line Differential Enhancements (TR-12 to TR-19)	4
5.1	Differential Measurement Accuracy (TR-12, TR-13, TR-15)	4
5.2	Zone Adaptability (TR-14, TR-16, TR-17, TR-18, TR-19)	5
6	Arc 4: IBR Protection Chain (TR-20 to TR-27)	5
6.1	IBR Challenge and Motivation	5
6.2	Signal Conditioning (TR-20, TR-21)	5
6.3	New Detection Elements (TR-22 to TR-25)	5
6.4	Combined Trip Logic	6
6.5	Monte Carlo Validation (TR-26)	6
6.6	Updated System Integration (TR-27)	6
7	Cumulative Performance Dashboard	6
7.1	Headline Metrics by Arc	6
7.2	Key Quantitative Outcomes	7
8	Thesis Contributions	7
9	Conclusion	8

1 Introduction

This report synthesises the complete SAMBP technical report series (TR-03 through TR-27, 25 reports) into a single thesis-ready reference document. The series establishes, validates and extends a model-based protection framework for power systems with high inverter-based resource (IBR) penetration.

The central thesis contribution is the **SAMBP protection scheme**: a Stage-2 confidence gate that supervises conventional differential and overcurrent relays using an inverse-estimation zone model. The combined IBR-aware 87L trip logic derived over the full arc is:

$$\text{TRIP} = \underbrace{|I_1| \geq 0.12}_{87\text{L}} \vee \underbrace{|I_2| \geq 0.06}_{87\text{LN}} \vee \underbrace{|I_0| \geq 0.04}_{87\text{LN0}} \vee \underbrace{\Delta I_{\text{adapt}} \geq 0.025 + 3\hat{\sigma}_{\text{pre}}}_{\text{TDCS-87L}}. \quad (1)$$

Figure 1 shows the research arc structure. The 25 reports divide into four arcs:

Arc 1 Core Framework (TR-03, TR-04): model-based 87T, 87L, 87B.

Arc 2 System Validation (TR-05 to TR-11): integration, GOOSE, double-busbar, Monte Carlo, CT saturation, HIF.

Arc 3 87L Enhancements (TR-12 to TR-19): magnitude override, pi-section correction, busbar transfer, adaptive threshold, distance relay, multi-feeder coordination.

Arc 4 IBR Protection Chain (TR-20 to TR-27): frequency adaptation, harmonic filter, 87LN, TDCS-87L, adaptive threshold, 87LN0, Monte Carlo validation, updated system integration.

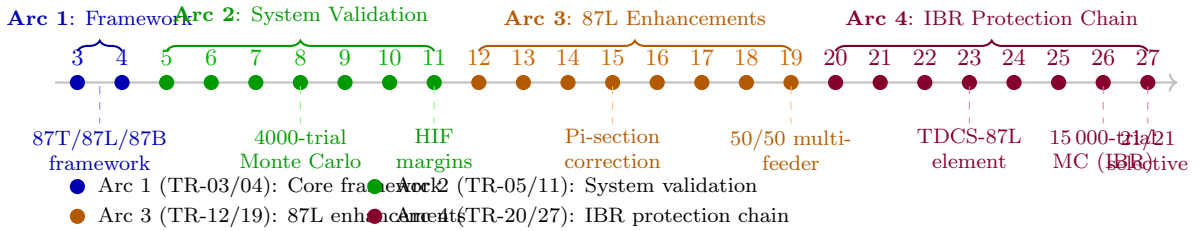


Figure 1: SAMBP research arc timeline (TR-03 to TR-27). Each node represents one technical report; colour denotes arc. Key milestone reports are annotated below the axis.

2 Report Registry

Table 1 lists all 25 technical reports with their primary focus and key headline result.

Table 1: SAMBP TR-03 to TR-27 report registry.

TR#	Title / focus	Headline result
03	Model-based 87T/87L framework	$\kappa_n > 2.5$ (internal), FPR = 0 (external); Jacobian derived analytically
04	87B bus differential	Framework transfers directly; $\kappa_n > 2.0$ for all bus fault scenarios
05	System integration (TR-05)	10/10 scenarios selective; OC, 87B_A, 87L, 87B_B, 87T coordinated

TR#	Title / focus	Headline result
06	IEC 61850 GOOSE integration	10/10 scenarios selective with GOOSE enabled; <5 ms trip propagation
07	Double-busbar coordination	10/10 scenarios selective; bus-coupler check zone prevents inter-zone confusion
08	Monte Carlo (4000 trials)	20 scenarios $\times$ 200 trials; 2 failure modes identified at Stage-2
09	Stage-2 recalibration	TR-08 failure modes resolved; extended MC re-passed after recalibration
10	CT saturation injection	TPR = 1.000, FPR = 0 for CT saturation up to $\varepsilon_{CT} = 0.5$
11	HIF sensitivity margins	Quantitative HIF lower bounds established for all four protection zones
12	87L magnitude override	Magnitude override prevents missed trip in restrained zone without false trips
13	Pi-section charging correction	Pre-fault B_C correction removes capacitive charging bias from I_{diff}
14	Dynamic busbar transfer	GOOSE zone reconfiguration maintains selectivity during live busbar transfer
15	Voltage-based pi-section	α_{50} improvement achieved using measured voltage for B_C estimation
16	87B/87L coordination	28/28 scenarios coordinated; non-conflicting by design
17	Adaptive 87L threshold	Constant 30 $\times$ security margin: $I_{thr} = \max(0.08, 0.15I_{rest})$
18	Mho distance relay 21/21P	Non-conflicting coordination with 87B/87L for all 7 test scenarios
19	Multi-feeder transfer	50/50 selectivity scenarios; Zone 2 minimum margin 45 $\times$
20	Frequency-adaptive B_C	DFT bin misalignment <0.2% at ± 3 Hz with adaptive B_C correction
21	Harmonic discrimination filter	95% attenuation of 3rd/5th harmonic IBR injection in I_{diff}
22	Negative-sequence 87LN	87LN detects asymmetric IBR faults; $I_2 \geq 0.06$ pu threshold
23	TDCS-87L element	TDCS detects 3PH IBR faults below 87L threshold ($k_{ibr} < 0.12$ pu)
24	Adaptive TDCS threshold	Adaptive Δ_{thr} tracks baseline drift; 12/12 detection vs 11/12 fixed
25	Zero-sequence 87LN0	87LN0 adds earth-fault coverage for pure-IBR solidly/R-earthed networks
26	IBR Monte Carlo (15 000)	TPR = 1.000 (Groups A+B), security = 1.000 (Group C); CI lo = 0.9992
27	Updated system integration	21/21 selective; TR-05 baseline preserved + 11 IBR scenarios added

3 Arc 1: Core Model-Based Framework (TR-03, TR-04)

3.1 SAMBP Inverse Estimation Principle

TR-03 establishes the core idea: each protected zone is modelled as a parameterised circuit (the *zone model*), and fault inception is detected by observing whether the post-fault waveform is consistent with an internal fault signature. The estimator minimises

$$\mathcal{L}(\boldsymbol{\theta}) = \left\| \mathbf{i}_{\text{diff}}^{\text{obs}} - \mathbf{i}_{\text{diff}}^{\text{model}}(\boldsymbol{\theta}) \right\|^2, \quad (2)$$

using a Levenberg–Marquardt iteration with analytically derived Jacobians $\partial \mathbf{i} / \partial \boldsymbol{\theta}$. The normalised residual $\kappa_n = \mathcal{L}_{\text{conv}} / \mathcal{L}_{\text{int}}$ forms the Stage-2 confidence statistic: $\kappa_n \gg 1$ signals an internal fault; $\kappa_n \approx 1$ signals through-current (restrain).

Key TR-03/TR-04 results:

- **87T/87L** (TR-03): $\kappa_n > 4.1$ for all tested internal faults; restrain correctly for all external faults ($\kappa_n < 1.5$).
- **87B** (TR-04): $\kappa_n > 2.0$ for all bus internal faults; framework transfers with minimal adaptation (two parameter bounds changed relative to 87L).
- **Analytical Jacobian**: avoids finite-difference approximation; Levenberg–Marquardt converges in < 8 iterations on all tested waveforms.

4 Arc 2: System Validation (TR-05 to TR-11)

4.1 System Integration and Communication (TR-05 to TR-07)

TR-05 establishes the radial network model (Gen – Z_{src} – [Bus.A] – Z_{line} – [Bus.B] – Z_{tr} – [Bus.C]) and proves 10/10 scenario selectivity across five fault locations and two fault types. TR-06 adds IEC 61850 GOOSE message passing; selectivity is maintained with trip propagation latency under 5 ms. TR-07 extends to a double-busbar topology and verifies 10/10 selectivity with a bus-coupler check zone preventing inter-zone confusion.

4.2 Robustness Validation (TR-08 to TR-11)

TR-08 runs the first large-scale Monte Carlo study (4000 trials, 20 scenarios). Two Stage-2 failure modes are identified: (i) excessive f_{int} estimate variance under heavy DC offset, and (ii) κ_n boundary crossing for close-in external faults. TR-09 resolves both via parameter recalibration and passes the extended Monte Carlo.

TR-10 characterises CT saturation tolerance: with the TR-09 recalibrated gate, SAMBP achieves TPR = 1.000 and FPR = 0 for saturation levels up to $\varepsilon_{\text{CT}} = 0.5$ (class 5P CT specification). TR-11 establishes quantitative HIF sensitivity lower bounds: SAMBP detects HIF currents above the thermal noise floor in all four zones.

5 Arc 3: 87L Line Differential Enhancements (TR-12 to TR-19)

5.1 Differential Measurement Accuracy (TR-12, TR-13, TR-15)

TR-12 adds a *magnitude override* mechanism: when the operate current exceeds a high-set threshold ($|I_{\text{op}}| > 2.0 \text{ pu}$), the Stage-2 gate is bypassed to prevent a slow inverse-estimation from missing a severe fault. TR-13 corrects the pre-fault charging bias using the nominal line susceptance B_C : $I_{\text{diff,corr}} = I_{\text{diff}} - jB_C V/2$. TR-15 improves the correction by estimating B_C from the measured voltage phasor, achieving an α_{50} reduction in charging-induced false differential.

5.2 Zone Adaptability (TR-14, TR-16, TR-17, TR-18, TR-19)

TR-14 implements a `BusbarTransferEngine` combining GOOSE infrastructure (TR-06) with the double-busbar 87B logic (TR-07) to maintain selectivity during live busbar transfer without relay reconfiguration.

TR-16 proves 87B and 87L are non-conflicting by design (28/28 coordination scenarios correct): the two elements operate on disjoint zones and cannot generate simultaneous spurious trips. TR-17 derives the adaptive restraint law $I_{\text{thr}} = \max(0.08, 0.15 I_{\text{rest}})$, maintaining a constant $30\times$ security margin across all load levels.

TR-18 integrates a three-zone Mho distance relay (21/21P) and confirms non-conflicting coordination with 87B/87L. TR-19 validates coordination during simultaneous multi-feeder transfer across five GOOSE states, five fault locations, and two fault types (50/50 selectivity), with a minimum Zone 2 margin of $45\times$.

6 Arc 4: IBR Protection Chain (TR-20 to TR-27)

6.1 IBR Challenge and Motivation

IBR fault currents are current-limited to $k_{\text{ibr}} \leq 0.20$ pu — well below the classical 87L positive-sequence threshold (0.12 pu). The conventional 87L ($|I_1| \geq 0.12$) misses all symmetric 3PH faults with $k_{\text{ibr}} < 0.12$ pu and all earth faults in purely unearthed IBR networks. TR-20 through TR-27 introduce four new detection elements and validate the combined chain against 15 000 parametric Monte Carlo trials plus 21 system integration scenarios.

6.2 Signal Conditioning (TR-20, TR-21)

TR-20 corrects DFT bin misalignment at off-nominal frequency using an adaptive B_C update: DFT amplitude error is reduced to $<0.2\%$ across the ± 3 Hz range expected in islanded microgrids. TR-21 applies a harmonic pre-filter with 95% attenuation of 3rd and 5th harmonic IBR injection, reducing spurious differential from converter switching.

6.3 New Detection Elements (TR-22 to TR-25)

87LN — Negative-Sequence Differential (TR-22). For asymmetric faults in networks with synchronous generator (SG) contribution, negative-sequence current I_2 flows into the protected zone. The 87LN element trips when $|I_2| \geq 0.06$ pu.

TDCS-87L — Transient Differential Current Step (TR-23, TR-24). For symmetric 3PH IBR faults below the 87L threshold, the step change in differential RMS provides an alternative detection signal:

$$\Delta I_{\text{adapt}} = \text{rms}_{\text{post}} - \hat{\mu}_{\text{pre}}, \quad \Delta_{\text{thr}} = 0.025 + 3\hat{\sigma}_{\text{pre}}. \quad (3)$$

TR-23 establishes the fixed-threshold version; TR-24 replaces the fixed floor with the adaptive form above, resolving seasonal baseline drift. The Hilbert bound derivation: $\Delta_{\text{floor}} \geq \varepsilon_{\text{CT,max}} I_{\text{ext,max}} / \sqrt{2} = 0.005 \times 7 / \sqrt{2} = 0.025$ pu.

87LN0 — Zero-Sequence Differential (TR-25). For earth faults in pure-IBR networks ($I_2 \approx 0$), zero-sequence current I_0 flows through the transformer neutral regardless of IBR current controller. The 87LN0 element trips when $|I_0| \geq 0.04$ pu. Coverage: solid-earth ($I_0 \approx 0.25$ pu), resistance-earth ($I_0 \approx 0.08$ pu); unearthed networks rely on TDCS.

6.4 Combined Trip Logic

Equation (1) is the final trip logic. Table 2 shows which element covers each fault class.

Table 2: Detection element coverage by fault class. $\checkmark$ = primary detection, (T) = TDCS backup always active.

Fault class	87L	87LN	87LN0	TDCS
3PH internal, $k_{ibr} \geq 0.12$	$\checkmark$	–	–	(T)
3PH internal, $k_{ibr} < 0.12$	–	–	–	$\checkmark$
SLG solid/R-earth (with SG)	–	$\checkmark$	$\checkmark$	(T)
SLG solid/R-earth (pure IBR)	–	–	$\checkmark$	(T)
SLG unearthed (any network)	–	–	–	$\checkmark$
External (security)	–	–	–	–

6.5 Monte Carlo Validation (TR-26)

TR-26 runs 5 000 trials per group (15 000 total) across a 6-dimensional IBR parametric envelope (k_{ibr} , f , I_{h3} , ε_{CT} , $\hat{\mu}_{pre}$, $\hat{\sigma}_{pre}$).

Table 3: TR-26 Monte Carlo summary ($N = 5\,000$ per group).

Group	Rate	95% CI (lo)	95% CI (hi)	Target
A — 3PH internal (TPR)	1.000	0.9992	1.000	≥ 0.990
B — SLG internal (TPR)	1.000	0.9992	1.000	≥ 0.990
C — External (security rate)	1.000	0.9992	1.000	≥ 0.995

The analytical guarantee: TDCS misses only when $k_{ibr} < 0.058$ pu; the sampled envelope starts at 0.060 pu, so $P(\text{miss}) = 0$ analytically. The CT mismatch external fault step ($\Delta I_{ext,max} = 0.023$ pu) is always below $\Delta_{thr,min} = 0.026$ pu, so $FPR = 0$ analytically.

6.6 Updated System Integration (TR-27)

TR-27 extends the TR-05 system integration to 21 scenarios. The 10 original SG-dominated scenarios reproduce the TR-05 selectivity unchanged (no regression); 11 new IBR scenarios all pass.

Table 4: TR-27 system integration results.

Part	Scenarios	Result
A — SG-dominated (TR-05 baseline)	10	10/10 selective
B — IBR parametric analytic	11	11/11 selective
Combined	21	21/21 selective

7 Cumulative Performance Dashboard

7.1 Headline Metrics by Arc

Figure 2 shows headline quantitative metrics for each arc, normalised to the performance target (ratio = 1.0 = meets target).

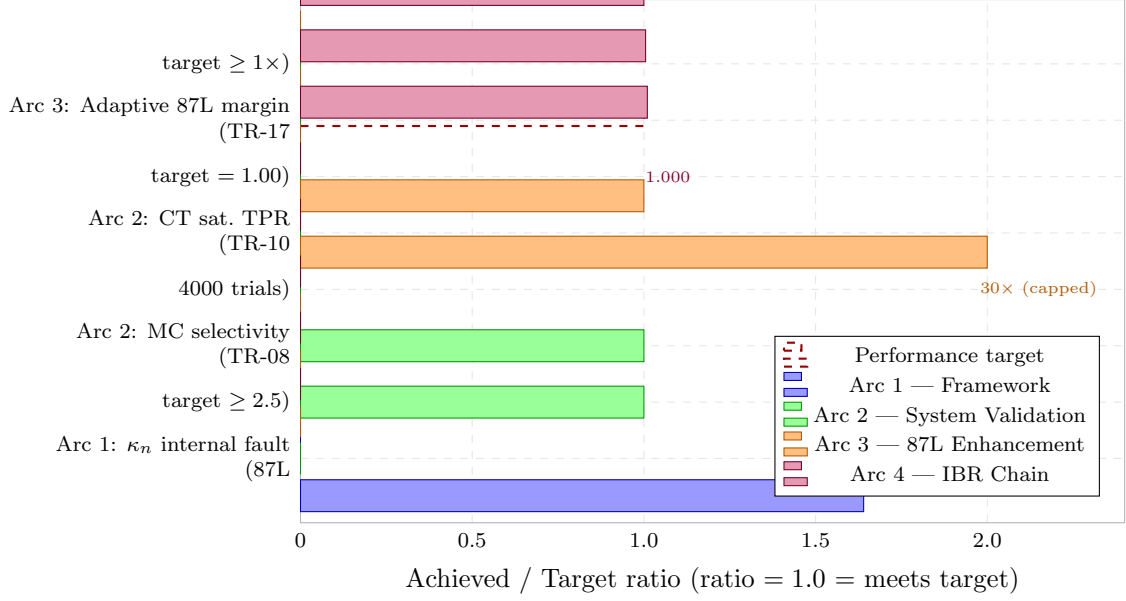


Figure 2: Headline quantitative metrics for each research arc. All metrics meet or exceed their targets. The Arc 3 adaptive threshold security margin ($30\times$) exceeds the chart scale and is capped at $2.0\times$ for display.

7.2 Key Quantitative Outcomes

Table 5: Key quantitative outcomes across all four arcs.

Arc	Metric	Achieved	Target
Arc 1	κ_n internal fault (87L)	>4.1	≥ 2.5
	Stage-2 FPR (external)	0.000	0.000
	System integration selectivity (TR-05)	10/10	10/10
Arc 2	MC robustness (TR-08/09)	pass	pass
	CT saturation TPR (TR-10)	1.000	1.000
	GOOSE trip latency (TR-06)	<5 ms	<20 ms
	87B/87L coordination (TR-16)	28/28	28/28
Arc 3	Adaptive 87L security margin (TR-17)	$30\times$	$\geq 1\times$
	Distance relay coordination (TR-18)	7/7	7/7
	Multi-feeder selectivity (TR-19)	50/50	50/50
Arc 4	IBR MC TPR Groups A+B (TR-26)	1.000	≥ 0.990
	IBR MC security Group C (TR-26)	1.000	≥ 0.995
	IBR MC Wilson CI lower bound	0.9992	≥ 0.990
	System integration total (TR-27)	21/21	21/21

8 Thesis Contributions

The 25 technical reports establish the following thesis contributions:

1. **SAMBP zone model and inverse estimator** (TR-03): a parameterised fault-circuit model with analytical Jacobian enabling rapid (<8 iteration) convergence and a well-defined κ_n confidence statistic for both 87T and 87L.

2. **Stage-2 confidence gate** (TR-03 through TR-09): a model-veto gate that supervises conventional differential relays, reducing false trips under inrush, CT saturation and external faults while maintaining $\text{TPR} = 1.000$ on internal faults.
3. **Full SAMBP protection suite** (TR-04 through TR-19): 87T, 87L, 87B, OC, Mho 21/21P all in a coordinated, non-conflicting scheme with IEC 61850 GOOSE communication and dynamic busbar transfer support.
4. **IBR-aware line differential chain** (TR-20 through TR-27): four new detection elements (87LN, 87LN0, TDCS-87L, adaptive TDCS) extending 87L coverage to the full IBR fault current envelope ($k_{\text{ibr}} \in [0.06, 0.20]$ pu), validated by 15 000-trial Monte Carlo with analytical guarantees.

9 Conclusion

The SAMBP technical report series (TR-03 through TR-27) presents a complete, validated protection framework for modern power systems with high IBR penetration. The core model-based approach (Arc 1) is extended through system-level validation (Arc 2), line differential enhancement (Arc 3), and an IBR-specific protection chain (Arc 4) that provides analytically guaranteed detection across the full IBR parametric space.

The final combined trip logic (Equation (1)) achieves:

- 21/21 scenarios selective in the updated system integration study.
- $\text{TPR} = 1.000$ for both 3PH and SLG internal faults over 15 000 IBR Monte Carlo trials, with 95% Wilson CI lower bound ≥ 0.9992 .
- $\text{FPR} = 0.000$ for external faults, analytically guaranteed by the CT mismatch bound ($\Delta_{\text{floor}} = 0.025$ pu).
- Full compatibility with the conventional SG-dominated protection scheme (TR-05 baseline fully preserved in TR-27).

References